COURSE PROJECT REVIEW 1

BACSE324 | Embedded Systems Architecture

Submission Template

Project Title

VisionGuard: Smart Cane for Visually Impaired People with Ultrasonic Obstacle Detection and Wet-Surface Warning

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Submission date	27 October 2026

1. Abstract (100 words)

People with vision loss mostly use a white cane, but it only finds obstacles by touching them and cannot warn about water, puddles or open drains. Our project, VisionGuard, is a smart cane built with an Arduino Nano. An HC-SR04 ultrasonic sensor checks the distance to objects in front, and a moisture probe at the bottom tip checks for water or wet ground. If the tip touches water, the buzzer sounds continuously. If an obstacle is closer than 50 cm, it beeps faster as the obstacle gets closer. Unlike most basic smart canes, it detects both obstacles and water, for under ₹900 in parts. In our tests the obstacle alert works on battery power; we are still calibrating the water threshold and plan to add vibration feedback.

2. Circuit and Block diagram

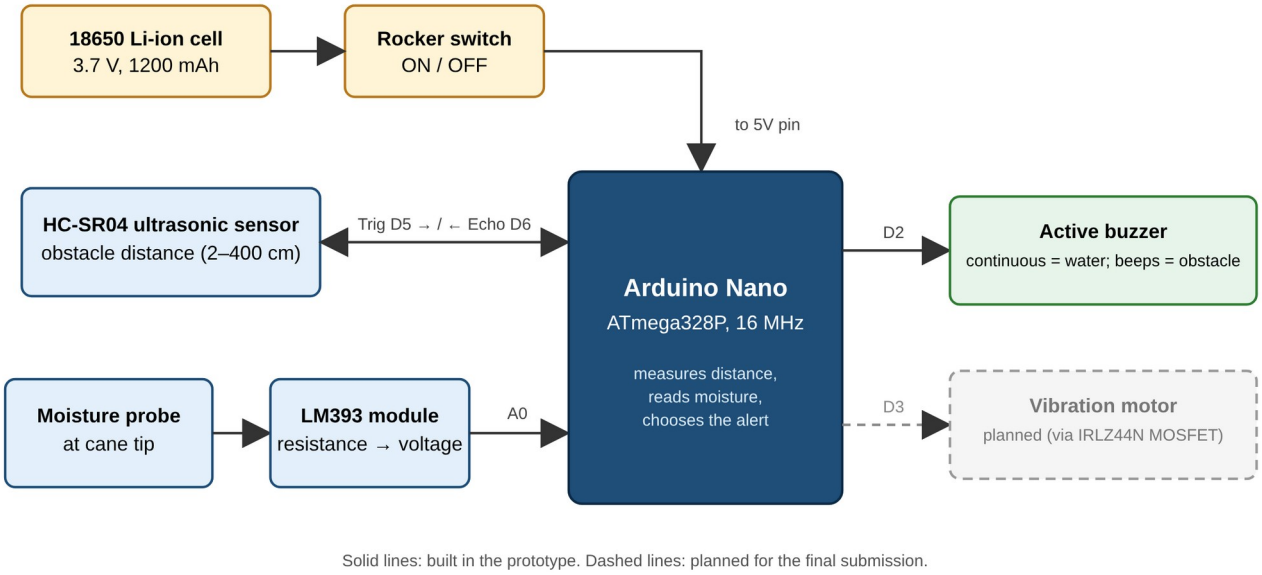


Figure 1. Block diagram of VisionGuard

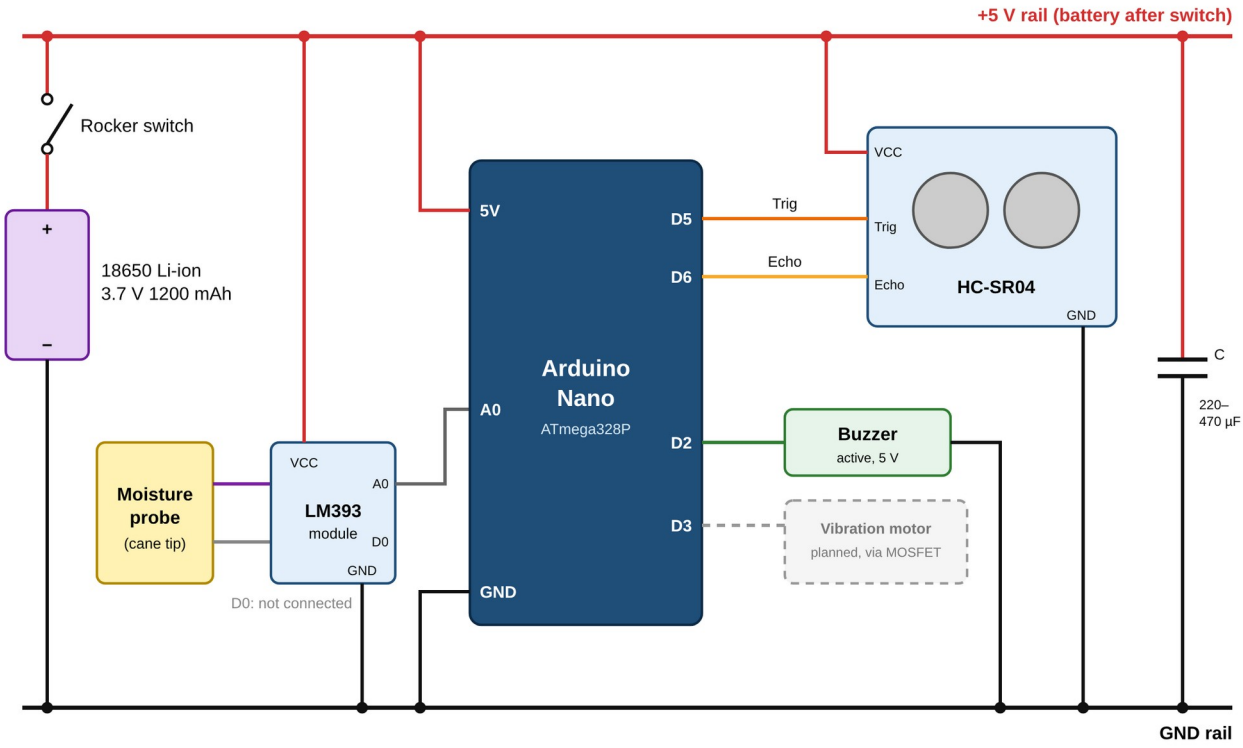


Figure 2. Circuit diagram. Pins as used in the firmware: Trig D5, Echo D6, moisture A0, buzzer D2; battery feeds the 5V pin through the rocker switch.

3. List of Components

List the main hardware and electronic components used in the prototype. Add or remove rows as required.

S. No.	Component	Specification / Model	Function in the project
1	Arduino Nano	ATmega328P, 16 MHz, 5 V logic, 10-bit ADC	Main controller: reads both sensors, runs the alert logic, drives the buzzer
2	Ultrasonic sensor	HC-SR04, 40 kHz, 2–400 cm range	Measures distance to obstacles in front of the cane (Trig D5, Echo D6)
3	Moisture probe	Fork-type resistive probe	Mounted at the cane tip; detects water or wet mud on contact
4	Moisture signal module	LM393 comparator board (A0 / D0 outputs)	Converts probe resistance into an analog voltage read on A0
5	Buzzer	Active buzzer, 5 V	Audio alert on D2: beeps for obstacles, continuous tone for water
6	Battery	18650 Li-ion cell, 3.7 V, 1200 mAh	Portable power source
7	Battery holder	18650 single-cell holder	Holds the cell on the cane
8	Rocker switch	SPST, 6 A	Power ON/OFF between battery and the 5V pin
9	Electrolytic capacitor	220 µF / 470 µF	Smooths the supply during buzzer switching
10	Cane body and wiring	PVC pipe, jumper wires, berg strips, single-strand wire	Mechanical structure and electrical interconnections

4. Bill of Materials (BOM)

Provide the estimated or actual cost of the components. Use the same currency throughout.

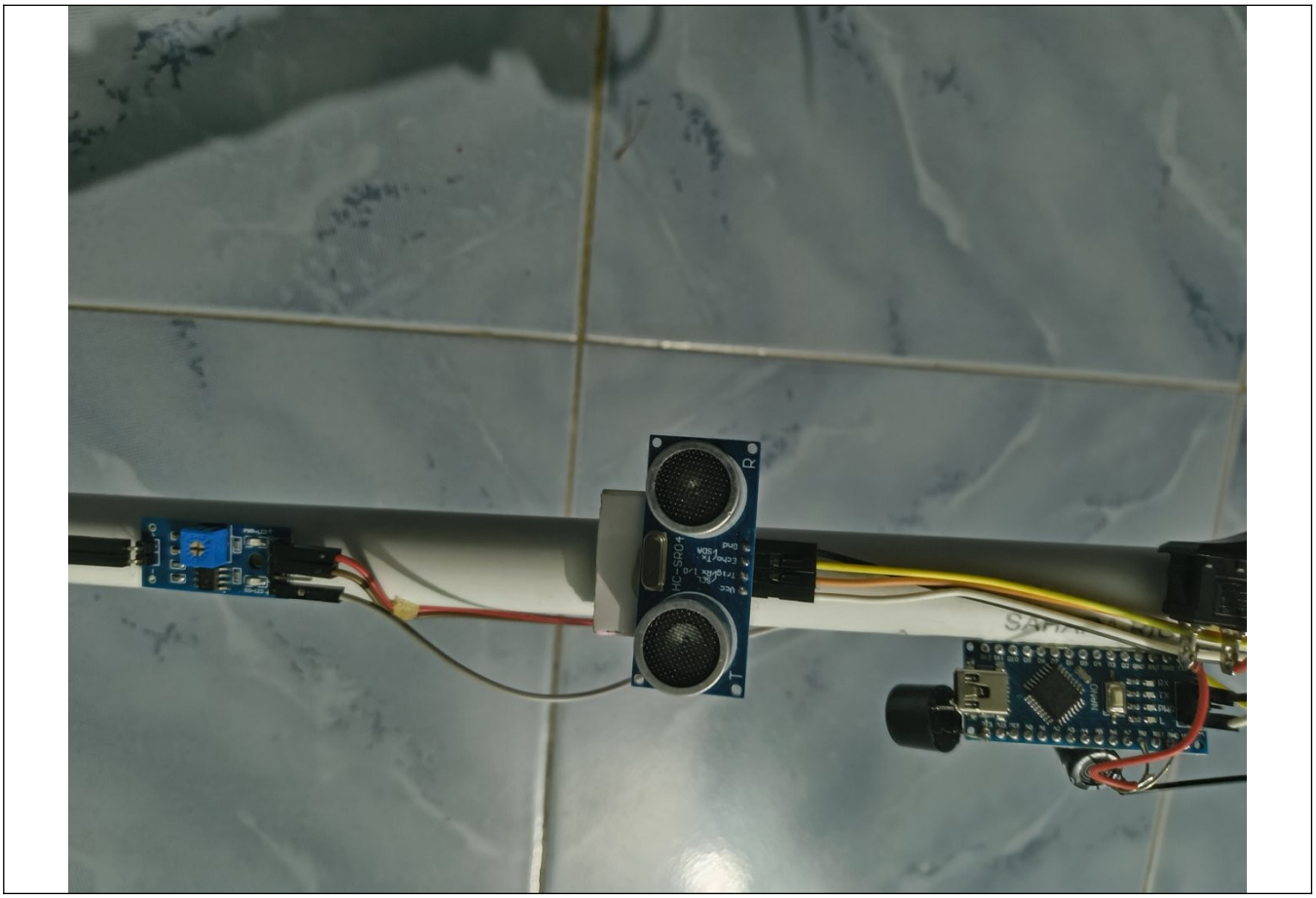
S. No.	Item / Component	Qty.	Unit cost (₹)	Total cost (₹)	Remarks
1	Arduino Nano (ATmega328P)	1	250.00	250.00	Vellore Robo Electronics
2	HC-SR04 ultrasonic sensor	1	105.00	105.00	Vellore Robo Electronics
3	Soil-moisture sensor (probe + LM393 module)	1	45.00	45.00	Vellore Robo Electronics
4	18650 Li-ion cell, 3.7 V 1200 mAh	2	65.00	130.00	One fitted, one spare
5	18650 cell holder	1	30.00	30.00	Billed as 2-cell; 1-cell holder fitted
6	Rocker switch, 6 A	1	15.00	15.00	Vellore Robo Electronics
7	Electrolytic capacitors (470 µF ×2, 220 µF ×2)	4	–	45.00	Supply filtering
8	Jumper wires (F-F, M-M, M-F; 10 each)	3 sets	20.00	60.00	Vellore Robo Electronics
9	Berg strips (2 female, 3 male)	5	10.00	50.00	Vellore Robo Electronics
10	Single-strand wire	3	12.00	36.00	Vellore Robo Electronics
11	Nano USB cable	1	40.00	40.00	For programming

S. No.	Item / Component	Qty.	Unit cost (₹)	Total cost (₹)	Remarks
12	Active buzzer, 5 V	1	20.00	20.00	Estimated; not on bill
13	PVC pipe, hot glue, cable ties	–	–	50.00	Estimated; cane body
14	Tools: soldering iron, solder, flux, breadboard	4	–	660.00	Not part of the cane
15	Parts for planned features: IRLZ44N MOSFET, 1 A fuse + holder, JST connector, resistor box, 0.1 µF capacitors	–	–	269.00	For Review 2 upgrades
Estimated total project cost				**1,805.00**	Cane hardware (items 1–13): ₹876

Prices are from invoice Ref. 3293, Vellore Robo Electronics (07-10-2026), including 18% GST; the billed total is ₹1,735. Items 12–13 (₹70) are estimates.

4. Prototype Photograph (Working Hardware)

Insert a clear, recent photograph of the assembled working prototype below. The photograph should show the actual hardware (not only a simulation or circuit diagram). Ensure that the main components are visible; use a caption to briefly identify the prototype and, if useful, indicate its operating condition.



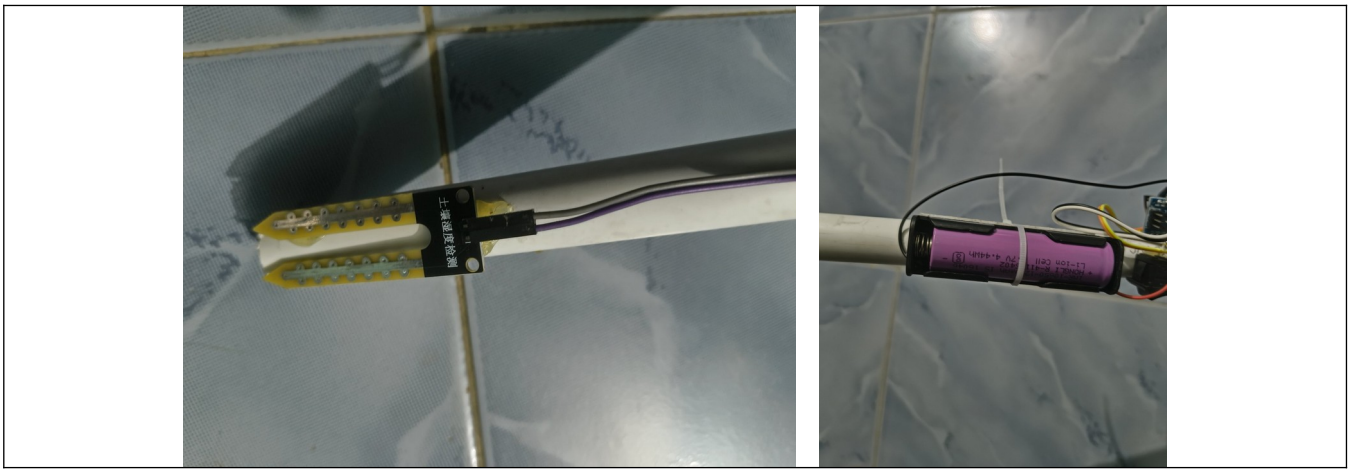


Figure 3. Our VisionGuard prototype on a PVC pipe. Top: LM393 module, HC-SR04 sensor, and Arduino Nano with buzzer and switch. Bottom left: moisture probe at the tip. Bottom right: 18650 battery in its holder. The cane gives obstacle alerts when running on the battery.

5. Current Status at Review 1 and Potential improvements for final submission

Briefly indicate what has been completed and what remains.

- Completed: We studied the problem and first planned a smart shoe, then changed it to a cane, because the cane tip reaches water before the foot does and a sensor on the shaft stays steadier. We chose the parts, mounted the sensors, Arduino Nano, buzzer, battery and switch on a PVC pipe, and soldered the wiring. The code for distance measurement, moisture reading and alerts is written and uploaded.
- Demonstrated functionality / preliminary results: On battery power, the buzzer beeps when an obstacle comes within 50 cm, and it beeps faster as the obstacle moves closer. We checked this by moving a hand and a notebook towards the sensor. The water alert is in the code; we are noting the dry and wet readings on the Serial Monitor to set the threshold.
- Challenges or limitations: The single 3.7 V cell is connected straight to the 5V pin, so there is no regulation or low-battery warning. The ultrasonic sensor has a narrow beam (about 15°), so it can miss head-level or thin obstacles and steps going down. The probe only senses water when it touches it, and it will corrode over time. The buzzer may be hard to hear in traffic, and 50 cm is a short warning distance.
- Next steps before Review 2: Set the moisture threshold properly and increase the obstacle range to about 100 cm. Add a vibration motor using the IRLZ44N MOSFET. Add a 5 V boost converter, a charging module and the 1 A fuse. Average a few readings and keep at least 60 ms between pings (as the HC-SR04 datasheet suggests) to cut false alerts. Put the wiring in a proper enclosure and test the cane outdoors.